

8.6/8.7 Volume and Surface Area of Spheres Worksheet

MPM1D

Part A: Find the Surface Area of Each Sphere

1)



$$SA = 4\pi(3)^2$$
$$= 113.1 \text{ ft}^2$$

2)



$$SA = 4\pi(6)^2$$
$$= 452.4 \text{ cm}^2$$

3)



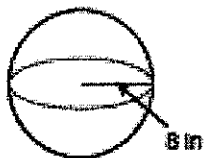
$$SA = 4\pi(7.8)^2$$
$$= 764.5 \text{ in}^2$$

4)



$$SA = 4\pi(7)^2$$
$$= 615.8 \text{ cm}^2$$

5)



$$SA = 4\pi(8)^2$$
$$= 804.2 \text{ in}^2$$

6)



$$SA = 4\pi(10)^2$$
$$= 1256.6 \text{ mi}^2$$

7) A sphere with a diameter of 6.2 in.

$$SA = 4\pi(3.1)^2$$
$$= 120.8 \text{ in}^2$$

8) A sphere with a radius of 10 mi.

$$SA = 4\pi(10)^2$$
$$= 1256.6 \text{ mi}^2$$

Part B: Find the Volume of Each Sphere

9)



$$V = \frac{4\pi(10.8)^3}{3}$$
$$= 5276.7 \text{ cm}^3$$

10)



$$V = \frac{4\pi(8)^3}{3}$$
$$= 2144.7 \text{ in}^3$$

11)



$$V = \frac{4\pi(6)^3}{3}$$
$$= 904.8 \text{ mi}^3$$

12)



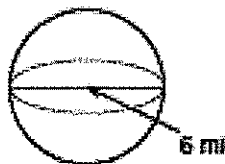
$$V = \frac{4\pi(7)^3}{3}$$
$$= 1436.8 \text{ in}^3$$

13)



$$V = \frac{4\pi(12)^3}{3}$$
$$= 7238.2 \text{ in}^3$$

14)



$$V = \frac{4\pi(3)^3}{3}$$
$$= 113.1 \text{ mi}^3$$

15) A sphere with a diameter of 2 m.

$$V = \frac{4\pi(1)^3}{3}$$
$$= 4.2 \text{ m}^3$$

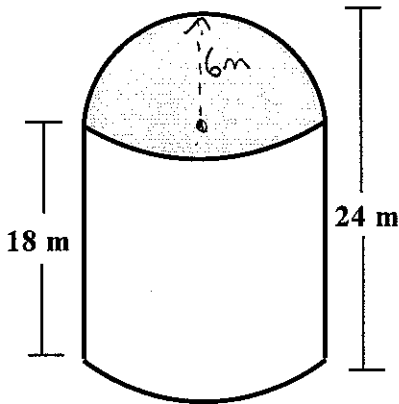
16) A sphere with a diameter of 10 ft.

$$V = \frac{4\pi(5)^3}{3}$$
$$= 523.6 \text{ ft}^3$$

Part C: Applications

17) Farmers store wheat in silos to protect the harvest from the weather.

- Determine how much paint is required to paint the surfaces of the silo that are exposed to the weather.
- Determine how much wheat the silo can hold?



1L covers 13m ²

$$\begin{aligned}
 \text{a) } SA_{\text{half sphere}} &= \frac{1}{2} 4\pi r^2 \\
 &= 2\pi r^2 \\
 &= 2\pi (6)^2 \\
 &= 226.2 \text{ m}^2
 \end{aligned}$$

$$\begin{aligned}
 SA_{\text{cylinder}} &= 2\pi rh \\
 &= 2\pi (6)(18) \\
 &= 678.6 \text{ m}^2
 \end{aligned}$$

$$\text{Paint needed} = 904.8 \text{ m}^2 \times \frac{1\text{L}}{13\text{m}^2} = 69.6 \text{ L}$$

$$\begin{aligned}
 \text{b) } V_{\text{cylinder}} &= \pi r^2 h \\
 &= \pi (6)^2 (18) \\
 &= 2035.8 \text{ m}^3
 \end{aligned}$$

$$\begin{aligned}
 \text{Total Volume} &= 2035.8 + 452.4 \\
 &= 2488.2 \text{ m}^3
 \end{aligned}$$

$$\begin{aligned}
 V_{\text{half sphere}} &= \left(\frac{1}{2}\right) \frac{4\pi (6)^3}{3} \\
 &= 452.4 \text{ m}^3
 \end{aligned}$$

18) A basketball has a diameter of 24.8 cm. $r = 12.4 \text{ cm} = 0.124 \text{ m}$

- How much leather is required to cover this ball, to the nearest tenth of a square centimeter?

$$\begin{aligned}
 SA &= 4\pi r^2 \\
 &= 4\pi (0.124)^2 \\
 &= 0.19 \text{ m}^2
 \end{aligned}$$

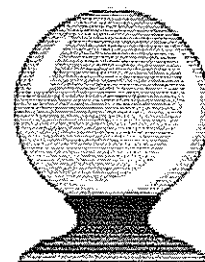


- If the leather costs \$28/m², what does it cost to cover the basketball?

$$\begin{aligned}
 C &= 28 \times 0.19 \\
 &= \$5.32
 \end{aligned}$$

19) Emily is placing a gazing ball in one of her customer's gardens. The ball has a diameter of $60\text{ cm} = 0.6\text{ m}$ and will be covered with reflective crystals. One jar of these crystals covers 1 m^2 .

a) Estimate the surface area to decide whether one jar of the crystals will cover the ball.



b) Calculate the surface area, to the nearest square centimeter.

$$SA = 4\pi(0.3)^2$$
$$= 1.13\text{ m}^2$$

This will require 2 jars.

20) Golf balls are stacked three high in a rectangular prism package. The diameter of one ball is 4.3 cm . What is the minimum amount of material needed to make the box?

$$h = 3 \times 4.3$$
$$= 12.9\text{ cm}$$

$$l \text{ and } w = 4.3$$

$$SA = 2lw + 2wh + 2lh$$

$$= 2(4.3)(4.3) + 2(4.3)(12.9) + 2(4.3)(12.9)$$

$$= 36.98 + 110.94 + 110.94$$

$$= 258.86\text{ cm}^2.$$



Answers: